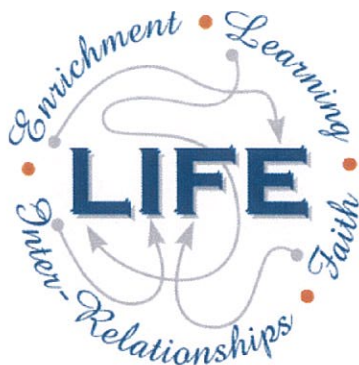




Holy Cross College

Semester 2 Examination 2016

Question/Answer Booklet

Year 10 Science

Please place your student identification label in this box

Student Name

SOLUTIONS

Student's Teacher

Time allowed for this paper

Reading time before commencing work: 10 minutes
Working time for paper: 110 minutes

Materials required/recommended for this paper

To be provided by the supervisor

This Question/Answer Booklet
Data Sheet

To be provided by the candidate

Standard items: pens, pencils, eraser, correction fluid, ruler, highlighters.

Special items: non-programmable calculators satisfying the conditions set by the School Curriculum and Standards Authority for this course.

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Suggested working time (minutes)	Marks available	Percentage of exam
Section One: Multiple Choice	56	56	40	56	35
Section Two: Short Answer	18	18	55	91	55
Section Three: Comprehension	9	9	15	17	10
				164	100

Instructions to candidates

1. The rules for the conduct of examinations at Holy Cross College are detailed in the College Examination Policy. Sitting this examination implies that you agree to abide by these rules.
2. Write your answers in this Question/Answer Booklet.
3. Working or reasoning should be clearly shown when calculating or estimating answers.
4. You must be careful to confine your responses to the specific questions asked and to follow any instructions that are specific to a particular question.
5. Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.
 - Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
 - Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.

SECTION 1 MULTIPLE CHOICE (55 marks)

Choose the best answer for each question in the Multiple Choice Question booklet.

Mark your selection with a cross.

1	(a)	(b)	(c)	(d)	31	(a)	(b)	(c)	(d)
2	(a)	(b)	(c)	(d)	32	(a)	(b)	(c)	(d)
3	(a)	(b)	(c)	(d)	33	(a)	(b)	(c)	(d)
4	(a)	(b)	(c)	(d)	34	(a)	(b)	(c)	(d)
5	(a)	(b)	(c)	(d)	35	(a)	(b)	(c)	(d)
6	(a)	(b)	(c)	(d)	36	(a)	(b)	(c)	(d)
7	(a)	(b)	(c)	(d)	37	(a)	(b)	(c)	(d)
8	(a)	(b)	(c)	(d)	38	(a)	(b)	(c)	(d)
9	(a)	(b)	(c)	(d)	39	(a)	(b)	(c)	(d)
10	(a)	(b)	(c)	(d)	40	(a)	(b)	(c)	(d)
11	(a)	(b)	(c)	(d)	41	(a)	(b)	(c)	(d)
12	(a)	(b)	(c)	(d)	42	(a)	(b)	(c)	(d)
13	(a)	(b)	(c)	(d)	43	(a)	(b)	(c)	(d)
14	(a)	(b)	(c)	(d)	44	(a)	(b)	(c)	(d)
15	(a)	(b)	(c)	(d)	45	(a)	(b)	(c)	(d)
16	(a)	(b)	(c)	(d)	46	(a)	(b)	(c)	(d)
17	(a)	(b)	(c)	(d)	47	(a)	(b)	(c)	(d)
18	(a)	(b)	(c)	(d)	48	(a)	(b)	(c)	(d)
19	(a)	(b)	(c)	(d)	49	(a)	(b)	(c)	(d)
20	(a)	(b)	(c)	(d)	50	(a)	(b)	(c)	(d)
21	(a)	(b)	(c)	(d)	51	(a)	(b)	(c)	(d)
22	(a)	(b)	(c)	(d)	52	(a)	(b)	(c)	(d)
23	(a)	(b)	(c)	(d)	53	(a)	(b)	(c)	(d)
24	(a)	(b)	(c)	(d)	54	(a)	(b)	(c)	(d)
25	(a)	(b)	(c)	(d)	55	(a)	(b)	(c)	(d)
26	(a)	(b)	(c)	(d)	56	(a)	(b)	(c)	(d)
27	(a)	(b)	(c)	(d)					
28	(a)	(b)	(c)	(d)					
29	(a)	(b)	(c)	(d)					
30	(a)	(b)	(c)	(d)					

SECTION 2 **SHORT ANSWER (87 marks)**

Attempt all questions. Write your answers in the spaces provided.

1. Classify the following elements as *alkali metals*, *transition metals*, *metalloid (semi-metals)*, *halogens* or *noble (inert) gases*. Use the periodic table on the data sheet provided.

K Ge He Cu Fr Cl Ag Ar

(4 marks)

Alkali Metal	Transition Metal	Metalloid (Semi-Metal)	Halogen	Noble (Inert) Gas
K Fr	Cu Ag	Ge	Cl	He Ar

$\left[\frac{1}{2} \text{ mark each} \right]$

2. Complete the following table showing the constituent particles of each element.

(4 marks)

Element	Protons	Neutrons	Electrons
$^{35}_{17}\text{Cl}$	17	18	17
$(^{32}_{16}\text{S})^{2-}$	16	16	18
$^{107}_{47}\text{Ag}$	47	60	47

$\left[\frac{1}{2} \text{ mark off each mistake} \right]$

3. An aluminium atom ($^{27}_{13}\text{Al}$) has the ability to form the Al^{3+} ion. It can then form ionic compounds with other ions such as Cl^{-} .

- (a) Draw a diagram to show the electron configuration in the *aluminium atom*.

(2 marks)



- (b) Explain what happens in order for the aluminium atom to become the aluminium ion Al^{3+} . (2 marks)

- The 3 valence electrons are donated. (1)
- The charge is $3+$ - 3 more protons than electrons. (1)

- (c) Why is the Al^{3+} ion so stable? (2 marks)

- It has a completely full outer shell. (1)
- It has the same electron configuration as Ne - the nearest inert gas. (1)

- (d) Explain how the "ionic bond" arises between the Al^{3+} ion and the Cl^{1-} ion. (2 marks)

- Al^{3+} is attracted to the Cl^{1-} ions. } (1)
- Opposite charges attract.
- Al donates electrons to Cl, making both atoms charged. (1)

4. Write the chemical formula for each of the following compounds.

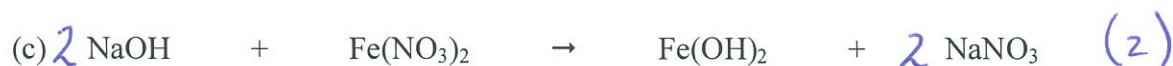
(7 marks)

NAME	FORMULA
potassium nitrate	KNO_3
magnesium phosphate	$\text{Mg}_3(\text{PO}_4)_2$
aluminium sulphate	$\text{Al}_2(\text{SO}_4)_3$
phosphorus tribromide	PBr_3
iron (III) iodide	FeI_3
barium acetate	$\text{Ba}(\text{CH}_3\text{COO})_2$
disulphur tetraiodide	S_2I_4

[1 mark each]

5. Balance the following equations.

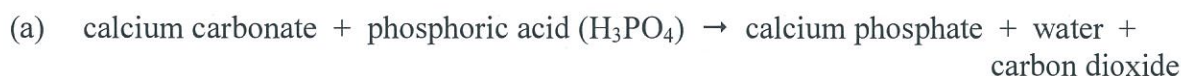
(6 marks)



[Be generous with errors!]

6. Write **fully balanced equations** for the following.

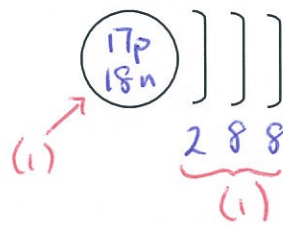
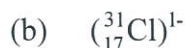
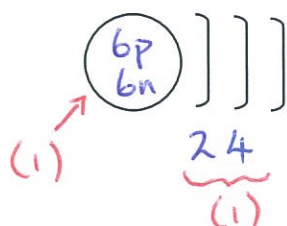
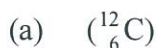
(4 marks)



[Correct formulae - 1 mark
Correct balancing - 1 mark]

Be generous!

7. Draw the electron configurations for the following. *Include the particles in the nucleus.* (4 marks)



8. Use the table of solubilities to determine whether a reaction occurs when the following solutions are mixed. Where a reaction occurs, write a **balanced equation**. Write "no reaction" if appropriate.

(Hint: It might be best to complete the word equation first to make it easier to write a balanced equation.) (4 marks)

- (a) Aluminium nitrate solution + sodium hydroxide solution.



[Subscripts are not important.]

[Correct products - 1 mark, correct balancing - 1 mark]

- (b) Barium nitrate solution + potassium sulphate solution.



9. A student makes two cups of tea and adds some sugar to each, allowing it to dissolve for 30 seconds. One cup has a teaspoon of sugar grains added while the other has a sugar cube added. (Assume the teaspoon of sugar and the sugar cube contain the same amount of sugar.)

- (a) Identify which cup of tea would be sweeter after 30 seconds. Justify your answer. (2 marks)

- Cup with the teaspoon of sugar. (1)
- Small particles increase the surface area of sugar that can dissolve. (1)

- (b) Describe two more things you could do to increase the rate the sugar dissolves in the tea. (2 marks)

- Stir the cup of tea. (1)
- Heat the tea / use hotter water. (1)

10. How long will it take to complete a trip by car from Perth to Kalgoorlie, a distance of 595 km, travelling at an average speed of 85 km/h? (3 marks)

$$\begin{aligned}
 v &= 85 \text{ km/h} \\
 d &= 595 \text{ km} \\
 t &= ?
 \end{aligned}
 \qquad
 \begin{aligned}
 v &= \frac{d}{t} \\
 \Rightarrow t &= \frac{d}{v} & (1) \\
 &= \frac{595}{85} & (1) \\
 &= \underline{7.0 \text{ hrs}} & (1)
 \end{aligned}$$

11. Vincent dropped a stone down a well and saw it hit the bottom 3.5 s later. Calculate the velocity of the stone at impact, given the acceleration due to gravity is 9.80 m/s^2 and the stone starts from rest. (3 marks)

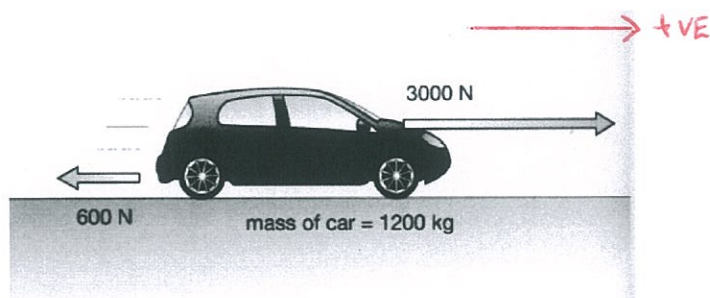
$$\begin{aligned}
 v &= ? \\
 u &= 0 \text{ m/s} \\
 a &= 9.80 \text{ m/s}^2 \\
 t &= 3.5 \text{ s}
 \end{aligned}
 \qquad
 \begin{aligned}
 &\downarrow +ve \\
 v &= u + at & (1) \\
 &= 0 + (9.80)(3.5) & (1) \\
 &= \underline{34.3 \text{ m/s}} & (1)
 \end{aligned}$$

12. Sean steps on the brake pedal of his car for 2.5 s. This applies a regarding force of 2100 N to slow his 750 kg car, which is travelling in an easterly direction. Calculate his car's deceleration. (3 marks)

Forwards is +ve.

$$\begin{aligned}
 F &= -2100 \text{ N} \\
 m &= 750 \text{ kg} \\
 a &= ?
 \end{aligned}
 \qquad
 \begin{aligned}
 F &= ma \\
 \Rightarrow a &= \frac{F}{m} & (1) \\
 &= \frac{-2100}{750} & (1) \\
 &= -2.8 \text{ m/s}^2 \\
 \therefore \text{Deceleration} &= \underline{2.8 \text{ m/s}^2 \text{ backwards}}. & (1)
 \end{aligned}$$

13. Consider the forces acting on the car in the diagram below.



- (a) Calculate the nett force acting on the body. (2 marks)

$$\begin{aligned}\Sigma F &= 3000 - 600 & (1) \\ &= \underline{2400 \text{ N forwards}} & (1)\end{aligned}$$

- (b) Calculate the acceleration of the body. (3 marks)

$$\begin{aligned}\Sigma F &= 2400 \text{ N} \\ m &= 1200 \text{ kg} \\ a &= ?\end{aligned}\quad \begin{aligned}\Sigma F &= ma \\ \Rightarrow a &= \frac{\Sigma F}{m} & (1) \\ &= \frac{2400}{1200} & (1) \\ &= \underline{2.0 \text{ m/s}^2 \text{ forwards}} & (1)\end{aligned}$$

14. Max has a mass of 57 kg on Earth. He is lucky enough to be selected for a junior astronaut programme and journeys to the International Space Station (ISS) for a 3-month stay.

- (a) Calculate Max's weight on Earth. (3 marks)

$$\begin{aligned}F_w &= ? \\ m &= 57 \text{ kg} \\ g &= 9.8 \text{ m/s}^2\end{aligned}\quad \begin{aligned}F_w &= mg & (1) \\ &= (57)(9.8) & (1) \\ &= \underline{558.6 \text{ N}} & (1)\end{aligned}$$

- (b) Max "floats" in space within the ISS. Is he weightless? Explain your answer. (2 marks)

- No (1)
- Gravity from the Earth is pulling on him. (1)

- (c) Is Max ever weightless in space? Explain your answer. (2 marks)

- No. (1)
- Gravity acts in all parts of the universe. (1)

15. A force of 36 N acts on a mass of 24 kg. What is the acceleration of the mass? (3 marks)

$$\begin{aligned}
 F &= 36 \text{ N} & F &= ma \\
 m &= 24 \text{ kg} & \Rightarrow a &= \frac{F}{m} & (1) \\
 a &= ? & & = \frac{36}{24} & (1) \\
 & & & = \underline{1.5 \text{ m/s}^2 \text{ forwards}} & (1)
 \end{aligned}$$

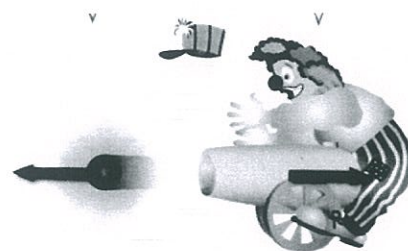
16. A mass of 28 kg is brought to rest by an average force of 98 N in 13 s. What is the initial velocity of the mass? (Hint - calculate the deceleration first.) (4 marks)

Forwards is +ve

$$\begin{aligned}
 F &= -98 \text{ N} & F &= ma \\
 m &= 28 \text{ kg} & \Rightarrow a &= \frac{F}{m} & (1) \\
 a &= ? & & = \frac{-98}{28} & (1) \\
 & & & = -3.5 \text{ m/s}^2 \text{ backwards} & (1) \\
 v &= 0 \text{ m/s} & v &= u + at \\
 u &= ? & \Rightarrow & 0 = u + (-3.5)(13) & (1) \\
 a &= -3.5 \text{ m/s}^2 & \Rightarrow & u = 45.5 \text{ m/s} & (1) \\
 t &= 13 \text{ s} & & &
 \end{aligned}$$

[Be generous!]

17. Look at the diagram showing a clown and a cannon at a circus. Analyse it and explain why the cannon hits the clown, using Newton's Laws in your explanation. (2 marks)



- ACTION - cannon ball moves forwards.
- REACTION - cannon moves backwards.

[1 mark each]

18. At the 2016 Singapore Grand Prix, Daniel Ricciardo accelerates quickly from the start and reaches 285 km/h (79.2 m/s) after 4.7 s. He is nearly at the first corner.

- (a) Calculate the average acceleration of the car.

(3 marks)

$$\begin{aligned}
 v &= 79.2 \text{ m/s} & a &= \frac{v-u}{t} & (1) \\
 u &= 0 \text{ m/s} & &= \frac{79.2-0}{4.7} & (1) \\
 a &= ? & &= 16.8 \text{ m/s}^2 \text{ forwards} & (1) \\
 t &= 4.7 \text{ s}
 \end{aligned}$$

- (b) Ricciardo is still travelling at 285 km/h (79.2 m/s) when he has to brake strongly to reduce the speed of the car to 110 km/h (30.6 m/s) to make it around the first corner. Given that the mass of the car is 780 kg and it takes 1.2 s to slow the car, calculate:

- (i) the deceleration of the car.

Forwards is +ve.

(3 marks)

$$\begin{aligned}
 v &= 30.6 \text{ m/s} & a &= \frac{v-u}{t} & (1) \\
 u &= 79.2 \text{ m/s} & &= \frac{30.6-79.2}{1.2} & (1) \\
 a &= ? & &= -40.5 \text{ m/s}^2 \\
 t &= 1.2 \text{ s} & & \therefore \text{Deceleration} = 40.5 \text{ m/s}^2 \text{ backwards} & (1)
 \end{aligned}$$

- (ii) the average force exerted by the brakes.

(3 marks)

$$F = ?$$

$$m = 780 \text{ kg}$$

$$a = -40.5 \text{ m/s}^2$$

Forwards is +ve

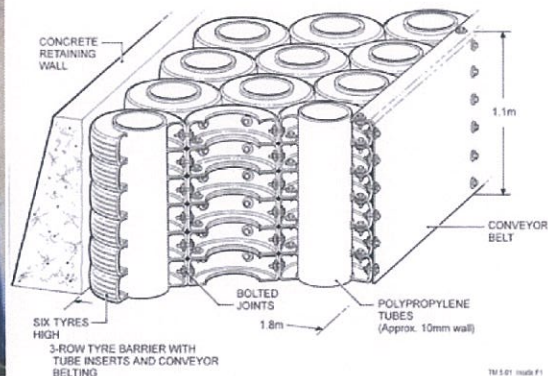
$$F = ma \quad (1)$$

$$= (780)(-40.5) \quad (1)$$

$$= -31,590 \text{ N}$$

$$\therefore \underline{F = 31,590 \text{ N backwards}} \quad (1)$$

- (c) At the start of the race, a car was hit and sent sideways into the rubber tyre wall, which acts as a safety device to protect the drivers.



- (i) Explain the Physics principle behind how the tyre wall acts as a safety barrier.

(2 marks)

- The tyres are able to collapse in a controlled way. (1)
- This slows the car over a longer time, decreasing the force on the car. (1)

- (ii) Tyre walls are around every corner on the track. When a car loses grip and starts to slide, it goes off the track **in a straight line**, even if it is skidding sideways. Using one of Newton's laws, explain why this happens.

(2 marks)

- Newton's First Law - the car has inertia. (1)
- Inertia makes the car move in a straight line - the car has a tendency to move off the curve. (1)

- (d) During the race, Ricciardo set a new race lap record of 110.041 s (1 min 50.041 s). Given the track is 5065 m long, calculate his average speed for the lap. (3 marks)

$$v = ?$$

$$d = 5065 \text{ m}$$

$$t = 110.041 \text{ s}$$

$$v = \frac{d}{t} \quad (1)$$

$$= \frac{5065}{110.041} \quad (1)$$

$$= \underline{46.0 \text{ m/s}} \quad (1)$$

SECTION 3

COMPREHENSION (17 marks)

Read the article and then answer the questions that follow. Write your answers in the spaces provided.

Mythconceptions—fry me to the moon

Dr Karl S. Kruszelnicki tackles the myths, curiosities and absurdities of everyday life.

One of the great myths about the space program that got us to the moon was that 'the only good thing to come out of it was Teflon'.

Teflon was in fact discovered, albeit accidentally, much earlier, on April 6, 1938, when a Du Pont chemist, Roy J Plunkett Jr. was looking for a new gas to use in refrigerators. One gas on his list was tetrafluoroethylene. Plunkett opened the valve of what was supposed to be a full cylinder of it, but none came out.

Most people would have just fetched another cylinder, but Plunkett was curious.

He weighed the cylinder and found that it weighed as much as a full cylinder of tetrafluoroethylene, so it wasn't empty. He then proved that the valve was indeed open by running a thin wire

into it. There was only one way left to work out what was going on, so he hacksawed the cylinder open, and found the gas had turned itself into a solid, white, slippery, waxy powder which, when analysed, revealed itself as polytetrafluoroethylene, today known as Teflon.

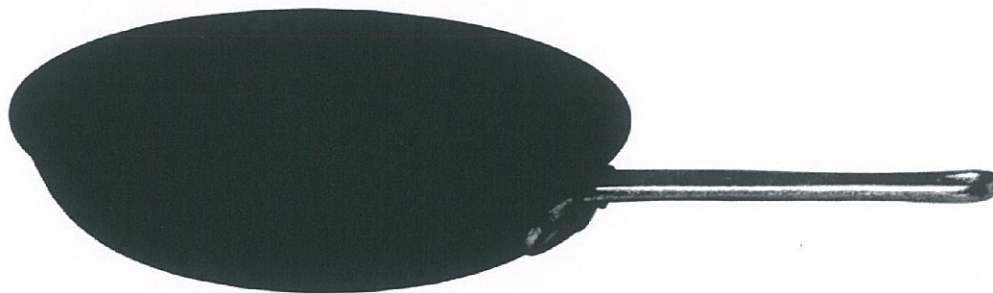
This powder proved to be the most inert plastic Plunkett had ever come across. It was unaffected by low heat. Nothing dissolved it. And it was more slippery than wet ice on wet ice. It was also a terrific electrical insulator. Sadly, it was very expensive to produce, so it had no commercial use.

During World War II, the Manhattan Project saw the first atom bombs made. As part of the manufacturing process, uranium had to be turned into uranium

hexafluoride. Unfortunately, this was an incredibly corrosive gas and would eat through metal, a major problem. But the Manhattan Project had money to burn, and once the US military heard about Teflon, it bought huge quantities of it to protect the machinery from the corrosive uranium hexafluoride.

But Teflon would also have applications in other important areas.

The human immune system tends to ignore Teflon, so it's used in artificial heart valves and aortas, as well as for artificial corneas and bones (for ears, chins, fingers, noses, skulls and various joints). Because it's tough enough to withstand the harsh conditions of outer space, Teflon is also used in spacesuits. And, of course, it's there in frying pans.



1. When was Teflon discovered and by whom?

- April 6 1938 (1)
- Roy Plunkett Junior (1)

(2 marks)

2. What is the correct chemical name of Teflon?

- polytetrafluoroethylene. (1)

(1 mark)

3. Describe what Teflon looks like.

- solid
 - slippery
 - white
 - waxy powder.
- [$\frac{1}{2}$ mark each]

(2 marks)

4. What properties make Teflon an extremely useful material?

- unaffected by low heat.
 - nothing dissolves it.
 - More slippery than wet ice.
 - Electrical insulator.
- [$\frac{1}{2}$ mark each]

(2 marks)

5. Teflon is said to be inert - what does this mean?

- won't react. (1)

(1 mark)

6. What research was the inventor doing when he discovered Teflon?

- Looking for a new gas to use in refrigerators. (1)

(1 mark)

7. What stopped Teflon from being commercially produced in the early days?

- Very expensive to produce. (1)

(1 mark)

8. Is Teflon a metal or a plastic? Give one piece of evidence.

- Plastic $(\frac{1}{2})$
- electrical insulator $(\frac{1}{2})$

(1 mark)

9. Complete the following table about the uses of Teflon.

LOCATION	USE	PROPERTY RELATED TO USES
Body	Heart valves Aorta Corneas Bones } Any one.	Immune system ignores it. (unreactive)
Space	space suit	Tough enough to withstand harsh conditions.
Kitchen	Frying pans	Slippery Unreactive } Either

(6 marks)

[1 mark each]

END OF EXAMINATION

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